

Target Look-Dev: ClayBound Production Standard

Objective:

Convert existing KFB 3D assets into tactile, handcrafted claymation models strictly without altering locked character designs.

Technical Pillars (The 'How'):

[Geometry] Softened macro-silhouettes via Modifier stacks. (Highlighted in Teal)

[Surface] Non-repetitive, tangent-mapped procedural/PBR hybrid texturing. (Highlighted in Amber)

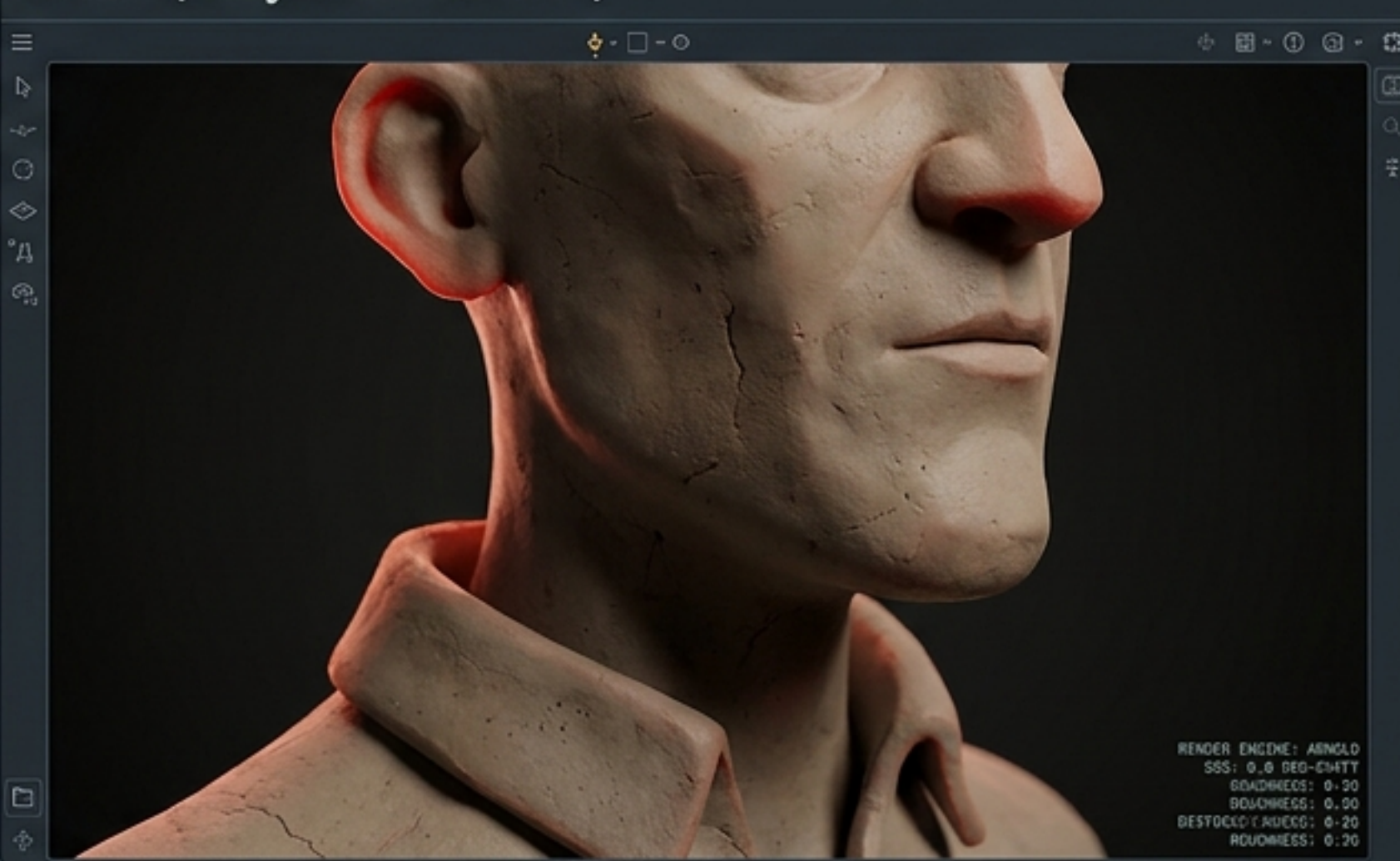
[Optics] High-roughness, dielectric shading with red-shifted Subsurface Scattering (SSS). (Highlighted in Blue)

[Motion] Frame-stepped procedural offset drivers (Stop-motion jitter). (Highlighted in Purple)

Before (Wireframe Topology)



After (ClayBound Render)



Surface Morphology: The Three Scales of Clay

Micro (Porosity)

Fine-grained starch structure, dry matte finish.

Implementation: High-frequency Normal Maps (T_Clay_MicroGrain_N) + High Roughness parameters.

Meso (Tool Marks)

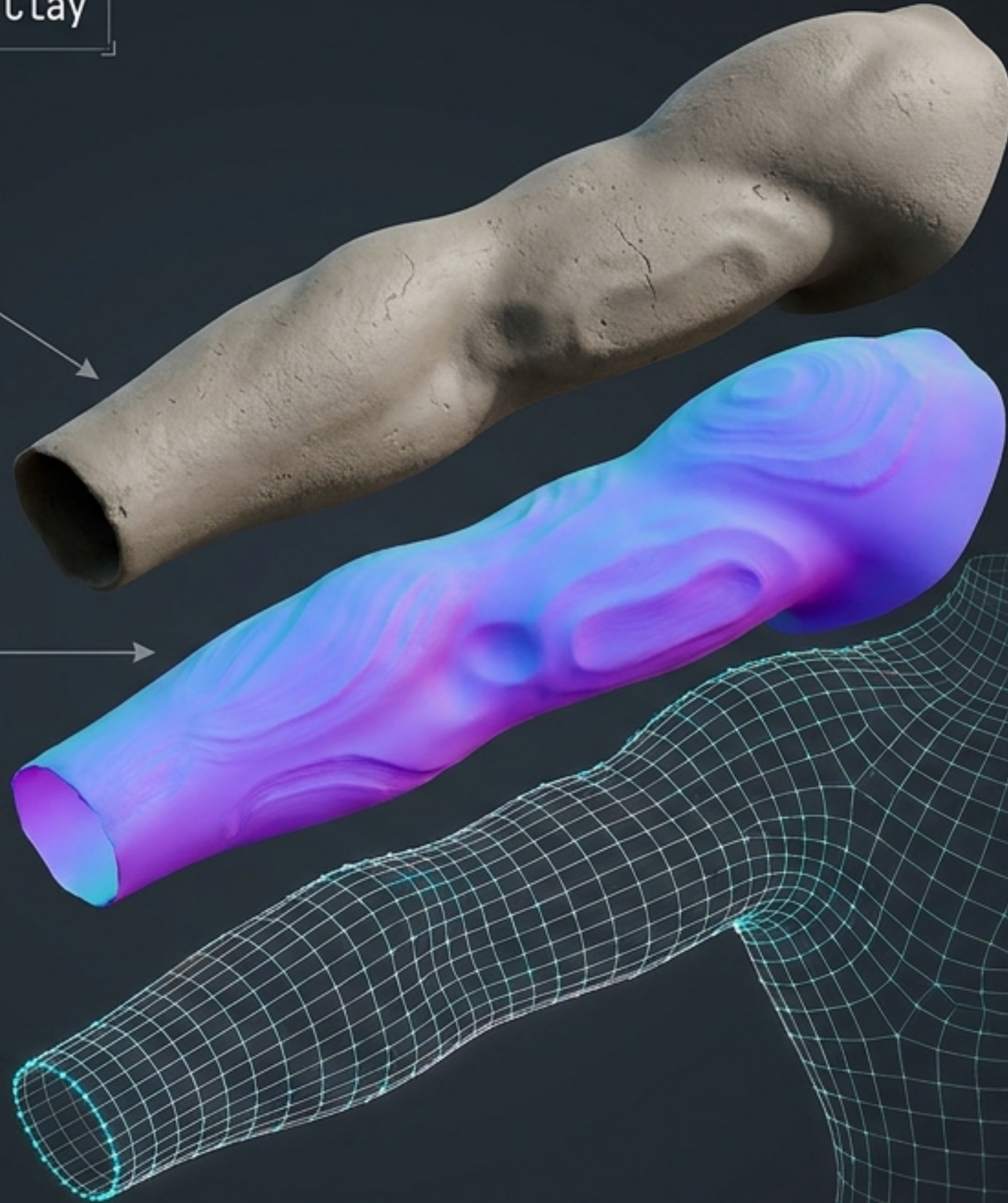
Broad contour variations, soft thumb indents, spatula scrapes.

Implementation: Low-frequency Normal Maps (T_Clay_MesoSculpt_N).

Macro (Form & Weight)

Rounded volumes, squishy joints, elimination of 90° polygons.

Implementation: Subdivision + Bevel Modifiers. (No manual sculpting required).



PROPERTIES

▼ MESH DATA

Vertices	14,592
Faces	14,560
Triangles	29,120

▼ MATERIAL SLOTS

 M_Clay_Master (Instance) ▼
Instance

▼ PARAMETER OVERRIDES

Roughness

0.55

Normal Strength (Meso)

1.0

Normal Strength (Micro)

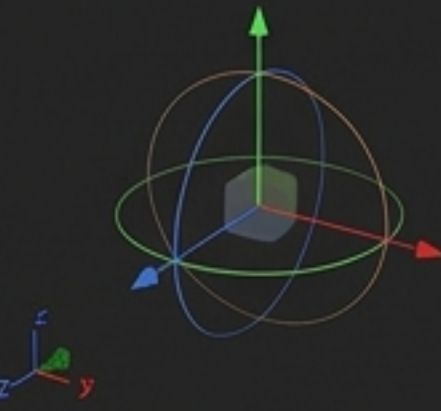
0.6

▼ TEXTURE INPUTS

 T_Clay_MesoSculpt_N
(2K, BC5)

 T_Clay_MicroGrain_N
(4K, BC5)

Render Queue



Render Queue

Preservation Directives: Modifying Surface, Retaining Identity

LOCKED (Do Not Alter)



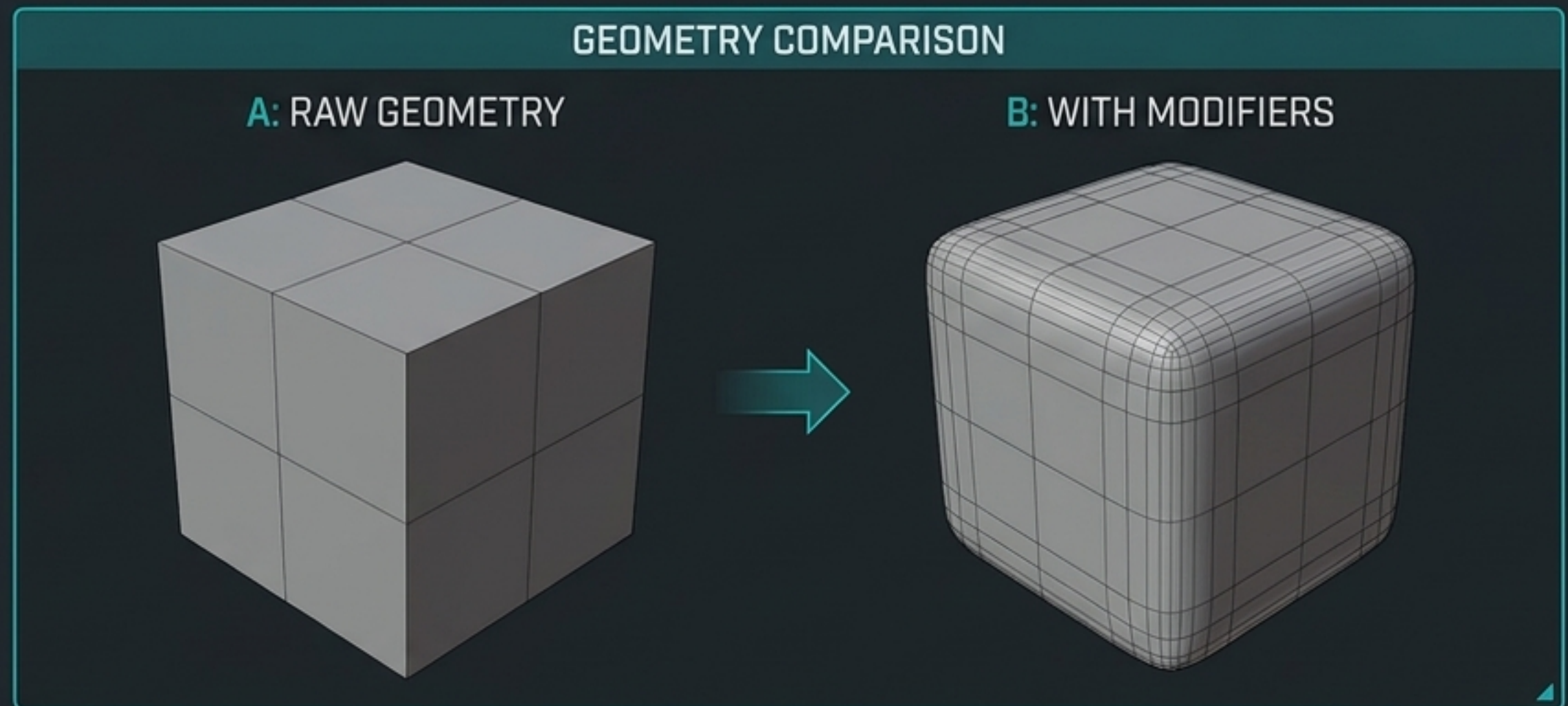
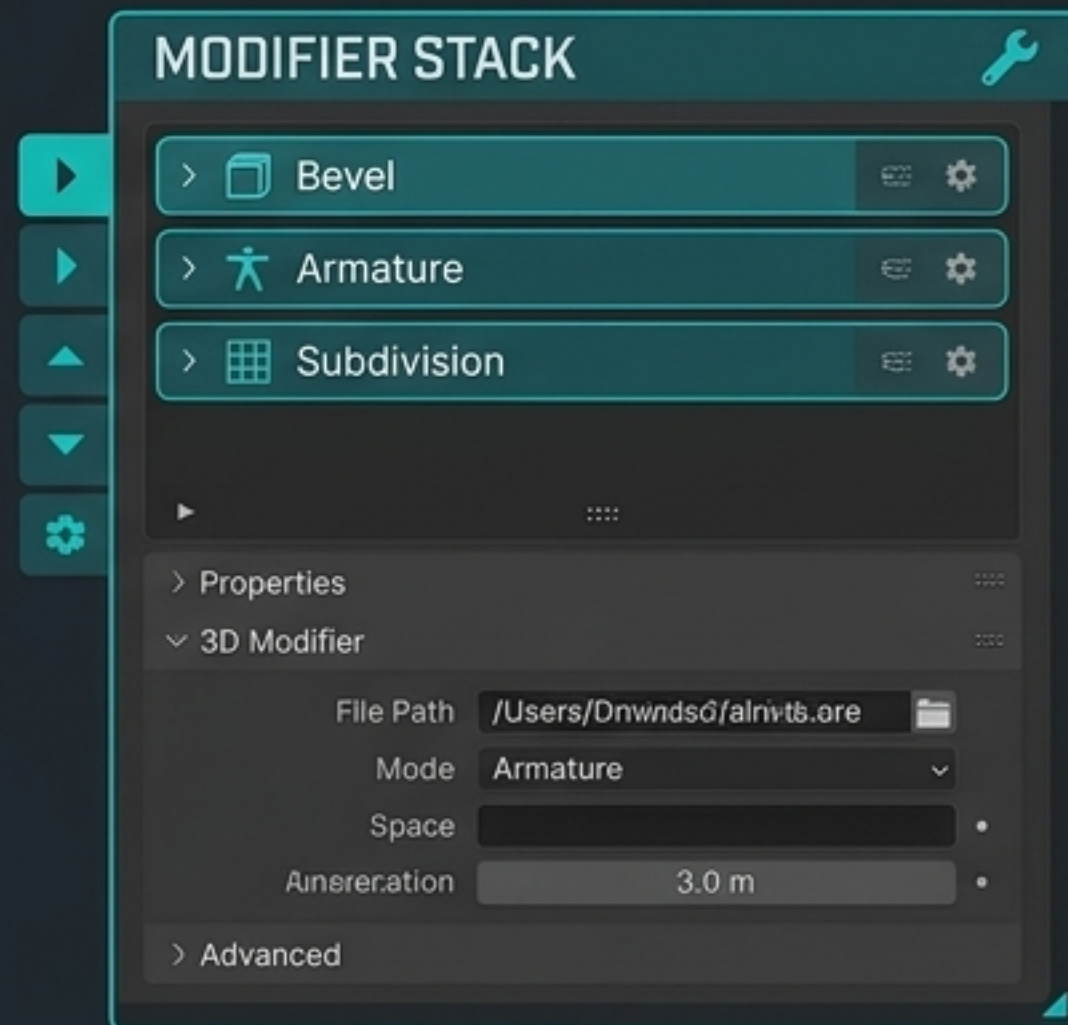
- Character Proportions & Silhouette Identity
- Skeletal Rig & Bone Hierarchy
- Base UV Unwraps (Must remain intact for seamless map mapping)
- Primary Base Colors (Brand hue identity must be preserved)

MUTABLE (Look-Dev Sandboxes)



- Edge Radii (Via procedural bevels/modifiers)
- Surface Displacement (Subtle, non-silhouette breaking)
- PBR Maps (Roughness, Normal frequencies)
- Lighting & Camera parameters

Macro Scale: Geometry Preparation Pipeline



Step 1: Scale Normalization

Ctrl+A -> Apply Scale. Critical for accurate SSS depth and Bevel radii calculation.

Step 2: Bevel Modifier

Width: 0.005m | Segments: 2 | Limit: Angle (30°)
Softens hard polygonal edges.

Step 4: Subdivision Surface

Render Levels: 2 | Viewport: 1
Ensures smooth light roll-off.

Step 3: Armature Deform

Place Bevel above Armature for static geometry, below if topology demands it for deformation.

Step 4: Subdivision Surface

Render Levels: 2 | Viewport: 1
Ensures smooth light roll-off.

Step 5: Shading

Enable Weighted Normals (Keep Sharp).

The PBR Texture Arsenal (Base Generation)

Golden Rule: No heavy fingerprints, no baked directional lighting, absolute seamless tiling.



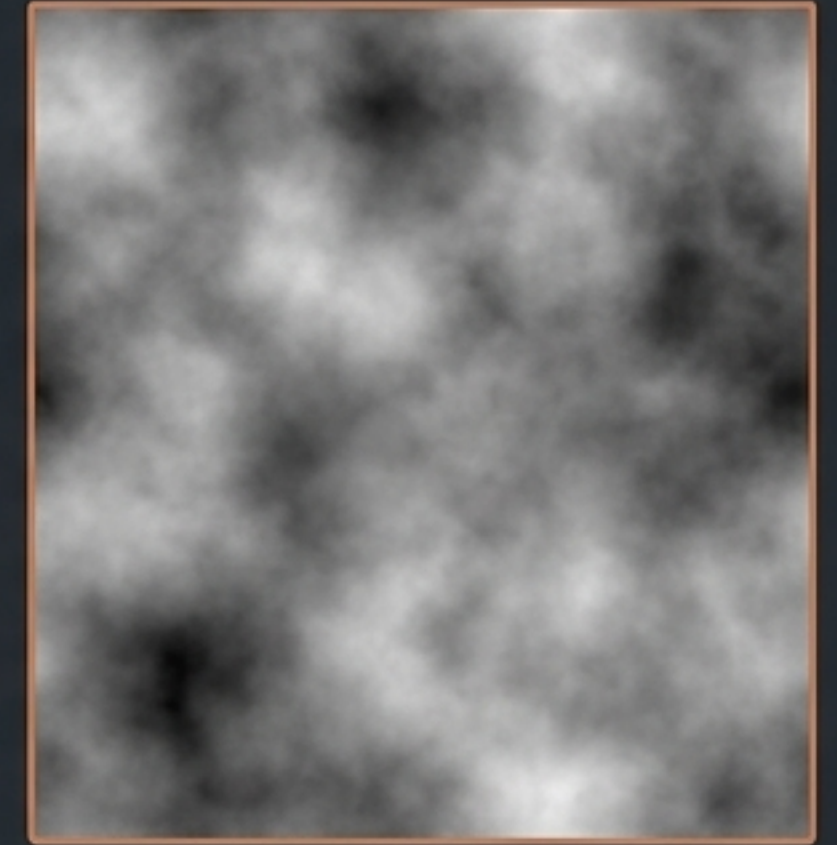
Swatch 1: Smooth Matte Clay
(Base skin)



Swatch 2: Softly Compressed Clay
(Broad tool strokes)



Swatch 3: Fine Porous Grain
(Micro-detail)



Swatch 4: Mottle Mask
(Low-frequency color breakup)

Generative Target / Prompt Template:

Seamless tileable matte modeling clay surface, extremely subtle organic microvariation, softly compressed handcrafted material, uniform neutral illumination, orthographic material sample, no border, no directional shadow.

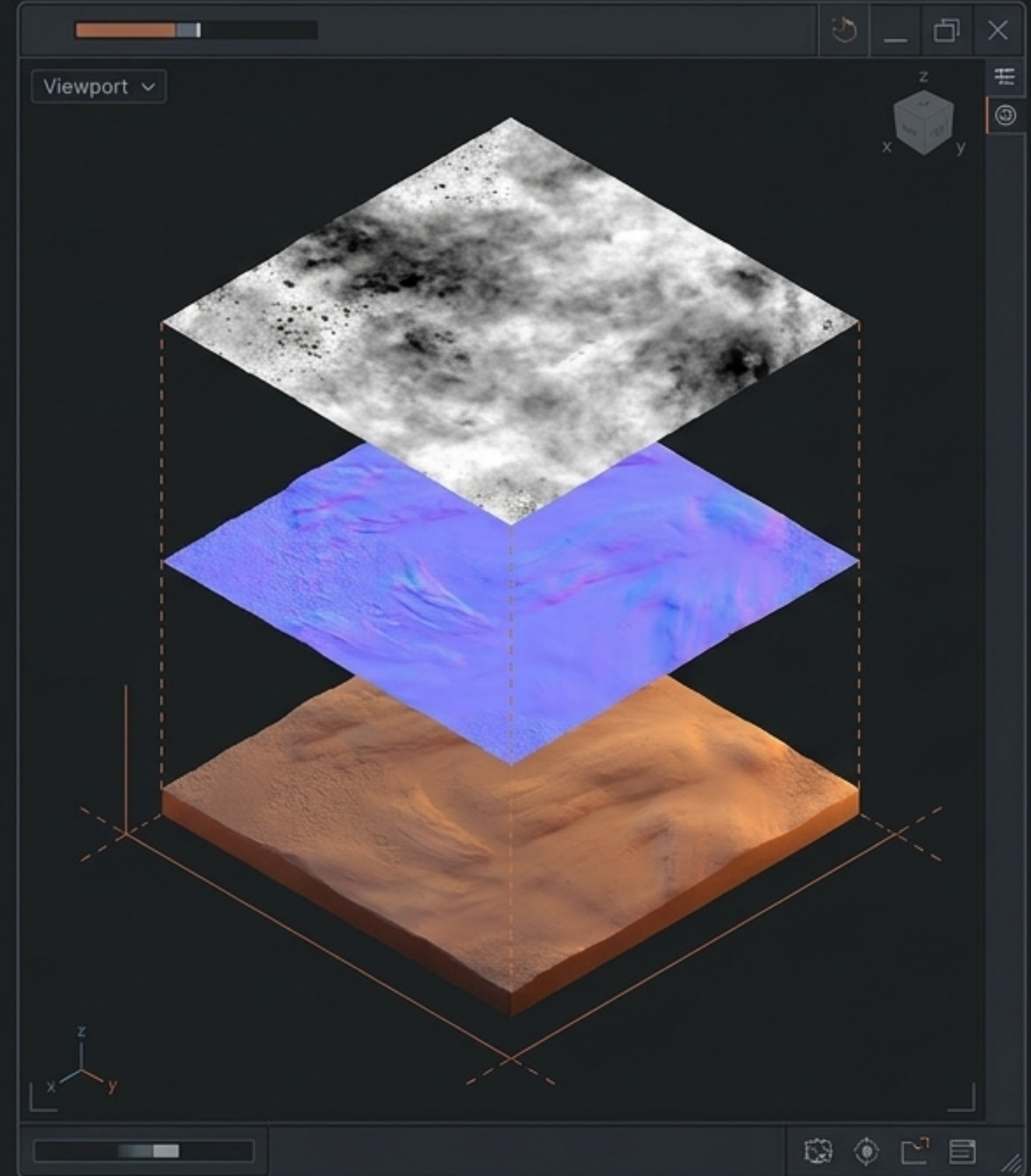
Map Technical Requirements & Color Spaces

T_Clay_MicroGrain_N	2048x2048 16-bit PNG/EXR
	Non-Color Space
	16-32x Tiling Target: Secondary Normal

T_Clay_MesoSculpt_N	2048x2048 16-bit PNG/EXR
	Non-Color Space
	2-4x Tiling Target: Primary Normal

T_Clay_MesoRoughness_R	1024x1024 8-bit PNG
	Non-Color Space
	2-4x Tiling Target: Roughness

T_Clay_MottleMask_M	1024x1024 8-bit PNG
	Non-Color Space
	1-2x Tiling Target: BaseColor Mix



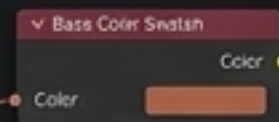
Shader Wiring: The ClayBound Master Node Group

Coordinates



Mapping: UV Texture Coordinate -> Mapping Node -> feeds all subsequent image textures.

Masks & Color



Color Chain: Base Color Swatch mixed with T_Clay_MottleMask_M via Mix Color (Soft Light).

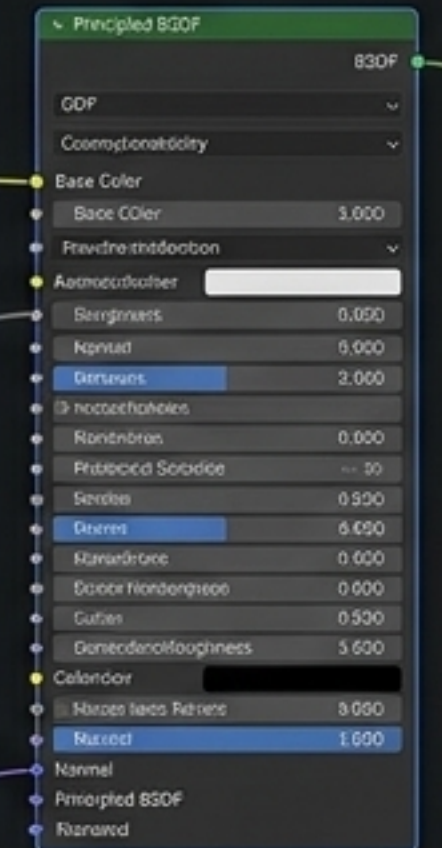
Normal Chain



Normal Chain Hierarchy (Crucial):

1. Bevel Shader Node (Radius 0.01m, 8 Samples) -> Vector input of:
2. T_Clay_MesoSculpt_N (Strength: 0.38) -> Vector input of:
3. T_Clay_MicroGrain_N (Strength: 0.15) -> BSDF Normal Input.

Principled BSDF



Physical Material Properties (PBR Ranges)

```
{  
  "Dielectric Rules": {  
    "Metallic":  
    "Coat Weight":  
    "_comment":  
  },  
  "Roughness": {  
    "_comment":  
  },  
  "Subsurface Scattering (The 'Glow')": {  
    "Method":  
    "Weight":  
  
    "Scale": "0.020m",  
    "_comment_scale":  
    "Radius (R,G,B)": [1.000, 0.350, 0.200],  
  },  
  "Sheen": {  
    "Weight": 0.150,  
  },  
  "Refraction": {  
    "Specular IOR Level": 0.450,  
    "IOR": 1.450  
  }  
}
```

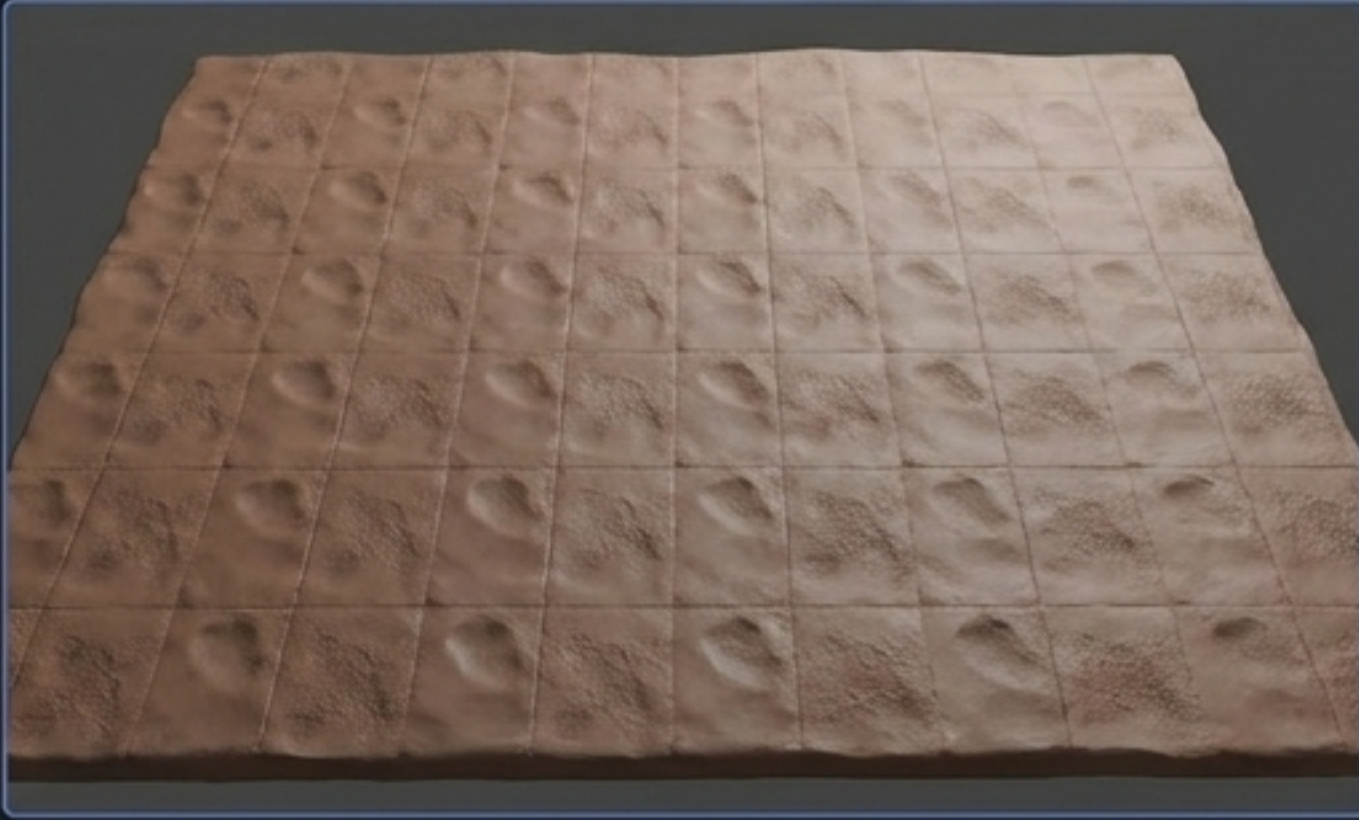
```
0.000,  
0.000,  
"_comment": "Strictly no glossy glaze/plastic look"  
  
"Range": "0.650 - 0.850",  
"_comment": "Matte finish, broad specular highlights"  
  
"Random Walk",  
0.250,  
"_comment": "Provides glow without jelly-like transparency",  
"Scale": "0.020m",  
"_comment_scale": "Crucial: adjust based on character world-scale",  
"Radius (R,G,B)": [1.000, 0.350, 0.200],  
"_comment_radius": "Red-shifted for warm internal light on thin edges"  
  
"_comment": "Micro-fiber edge softening"
```


Breaking the Grid: Scale & Tiling Calibration

The Problem: Uniform UV scale reveals map tiling on large/duplicated KFB assets.

The Solution: Multi-frequency blending and randomized coordinate offsets.

Left Render: Uniform Tiling (Problem)



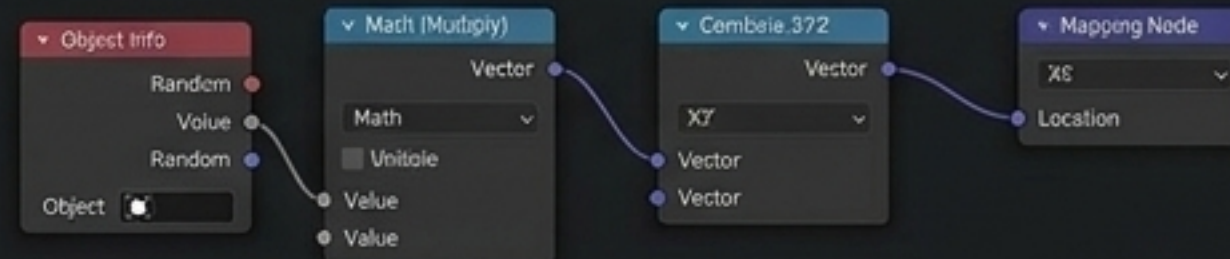
Right Render: Randomized Tiling (Solution)



Implementation:

- Drive micro-grain (Scale: 24.0) independently of meso-sculpt (Scale: 3.0).

Node Driver:



Ensures duplicated props (e.g., character accessories) don't share identical noise patterns.

Zone Mapping: Preserving Character Identity

Primary Clay Body: Master Shader
(Full SSS, Meso+Micro normals).

Master_Shader_Clay
Subsurface: 1.0
Normals: Meso_Clay_N
 Micro_Clay_N

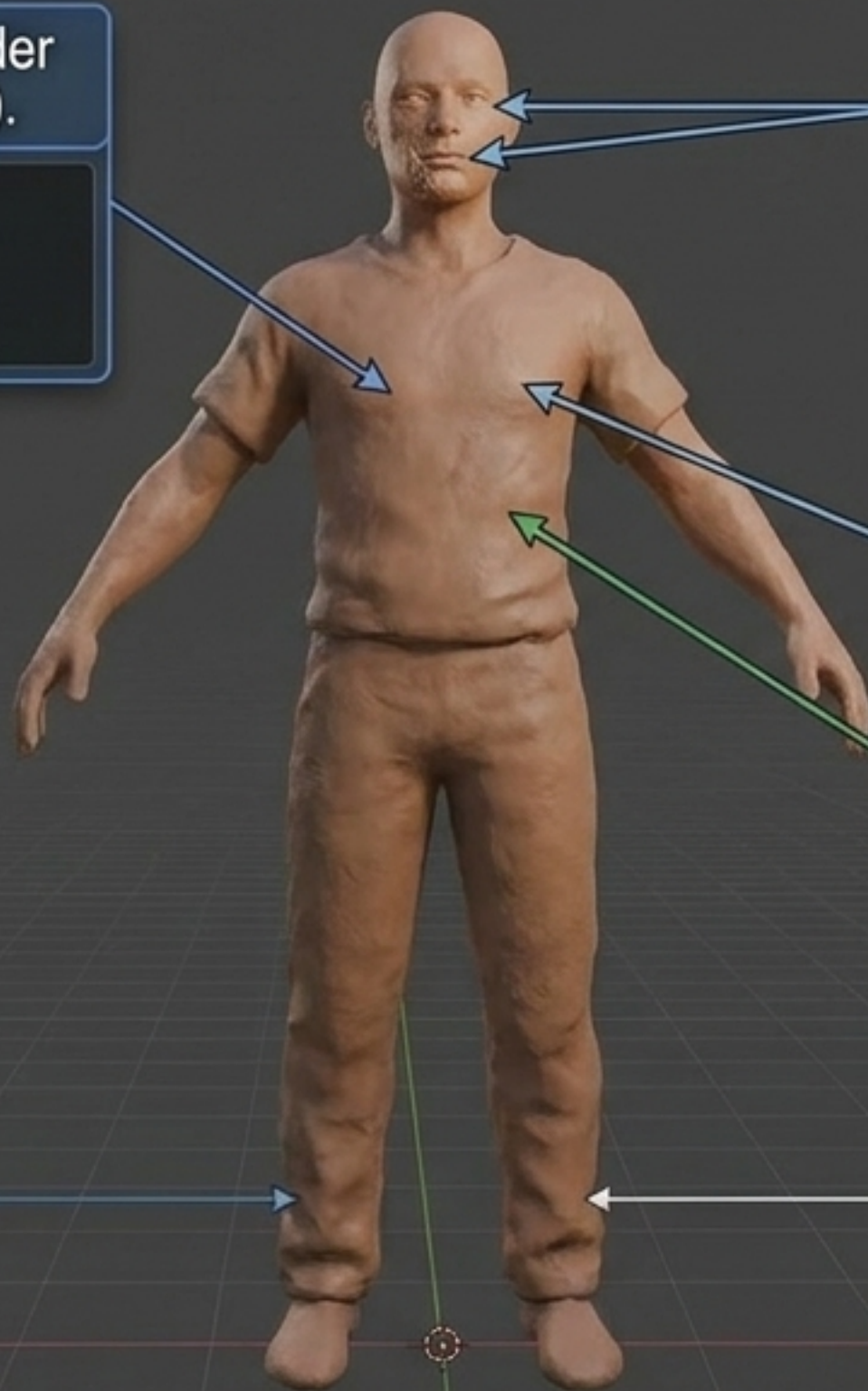
Face/Mouth/Eyes: Master Shader with
Subsurface Scale reduced by 50% to
prevent delicate facial features from
bleeding out or losing crisp definition.

Master_Shader_Clay (Instance)
Subsurface_Scale: 0.010m (50%)
Preservation: High

Execution Protocol:
Duplicate M_ClayBound_Master into
slot-specific variations (M_Clay_Skin,
M_Clay_Shirt). Change only the Base Color
swatch to match original KFB palettes.

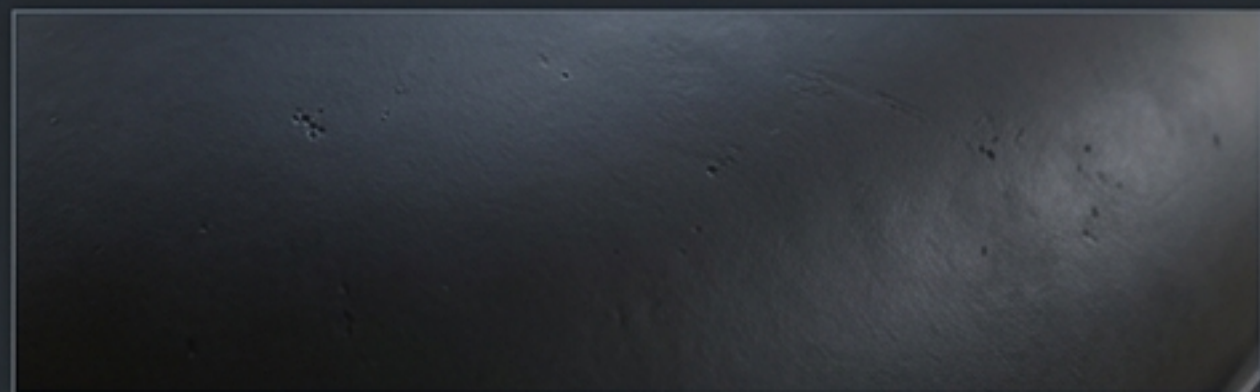
M_Clay_Shirt (from M_ClayBound_Master)
Base Color: [0.125, 0.350, 0.550] (KFB Blue)

M_Clay_Skin (from M_ClayBound_Master)
Base Color: [0.850, 0.700, 0.550] (KFB Skin)



Secondary Materials: Handcrafted Exceptions

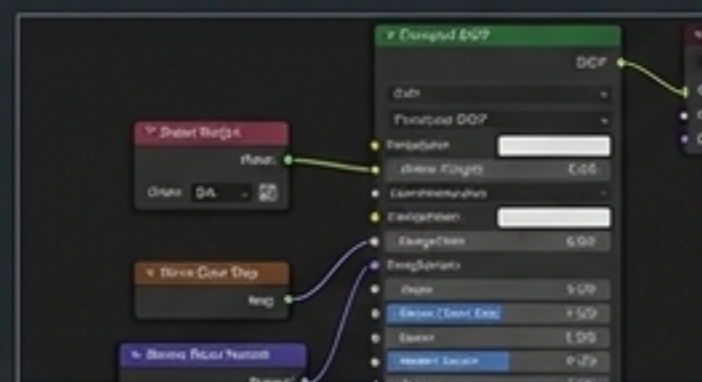
⚠ Note: Only deploy when justified by the existing KFB asset's locked design.



Matte Rubber (Shoes/Props):

Subsurface Weight -> 0.0

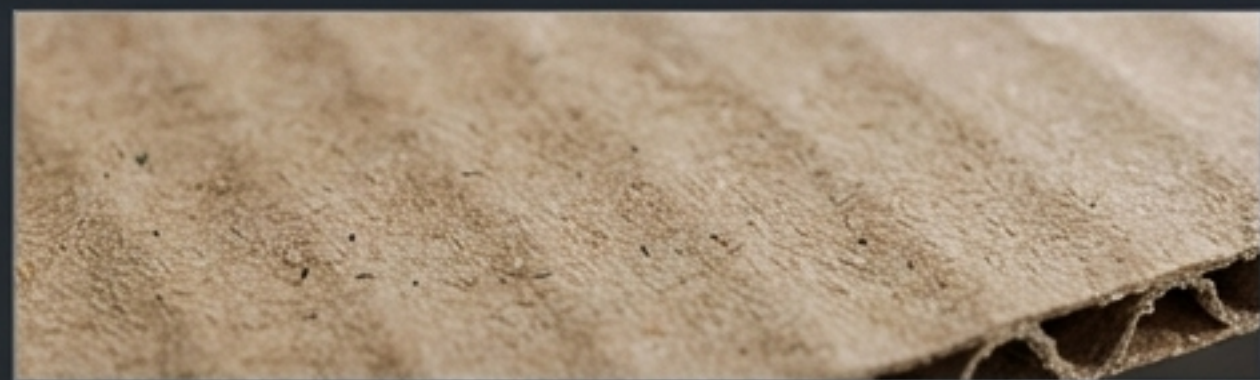
Roughness -> 0.90



Felt/Fabric (Capes/Hats):

Enable Sheen Weight -> 0.80

Replace micro-grain map with woven micro-normal.



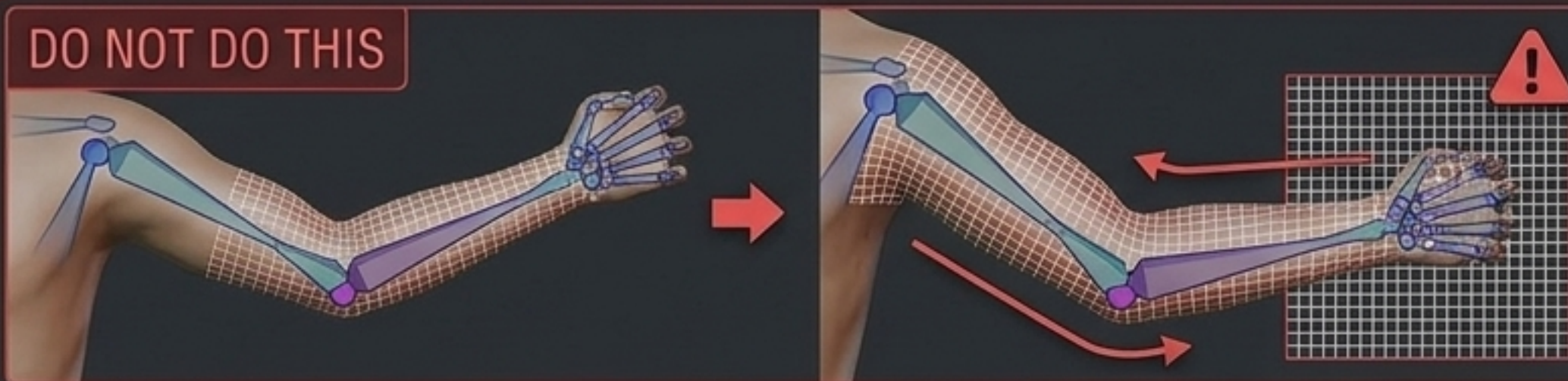
Cardboard/Paper (Accessories):

Subsurface Weight -> 0.0

Add Anisotropic -> 0.1 for subtle directional grain.

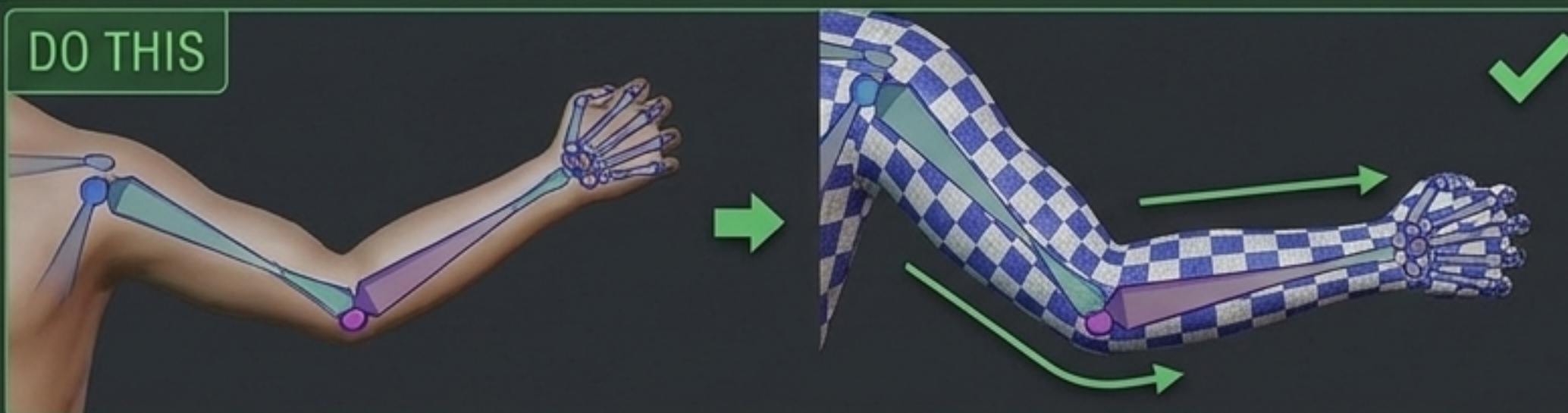
Motion Integrity: Preventing Texture Swimming

DO NOT DO THIS

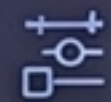


Using World/Object coordinates for procedural noise on skeletal meshes causes the texture to slide across the skin during bone deformation.

DO THIS

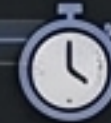


The Rule: All normal maps and procedural masks MUST use native Tangent-Space UV coordinates (Texture Coordinate: UV).

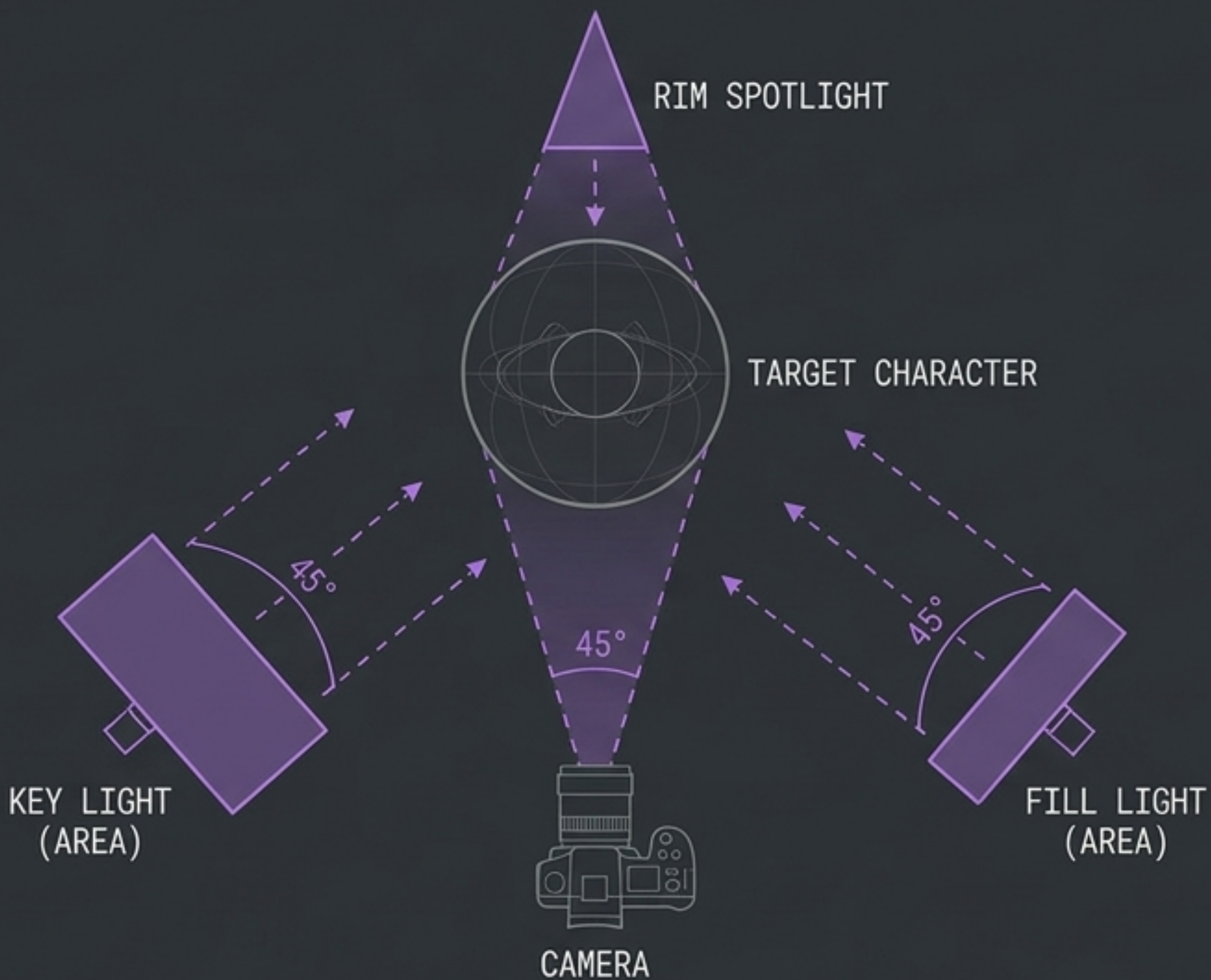
Exception (Stop-Motion Jitter): 

To emulate animator thumb-presses, use a quantized frame driver on the Mapping Node Location:

$$\text{Offset_X} = \text{floor}(\text{Frame} / N) * C$$

(e.g., updates noise offset discretely every 2 frames, holding static in between). 

The Presentation Rig: Studio Lighting & Optics



COLOR MANAGEMENT:

Transform: AgX | Look: Medium High Contrast.

(Crucial for warm SSS roll-off without highlight clipping).

CAMERA:

Focal Length: 85mm - 105mm
(Creates miniature macro lens perspective).

DoF: $f/4.0$ - $f/5.6$
(Miniature scale depth).

LIGHTING RIG:

Key: Area Light (1.5m), 250W, 5200K
(Warm main shadow).

Fill: Area Light (2.5m), 75W, 6500K
(Cool shadow lift).

Rim: Spot Light, 350W, 7000K (Crucial for accentuating SSS edge glow).

Asset Handoff: The Delivery Package

Folder Structure:

```
/KFB_CLAY_SYSTEM/  
├─ /textures/  
│   ├── T_Clay_MicroGrain_N.exr (16-bit Normal)  
│   └── T_Clay_MesoRoughness_R.png (8-bit Roughness)  
├─ /blender/  
│   └── KFB_Clay_Master.blend (Contains Master NodeGroup and Studio Lighting Rig)  
└─ /specs/  
    └── QA_CHECKLIST.md
```

GLB Export Protocol:

For real-time targets, bake procedural masks to BaseColor, export tangent normals, and bake displacement to mesh if topology allows.

Quality Assurance: The Evaluation Checklist

PASS CRITERIA [✓]

- Target KFB silhouette identity is preserved 1:1.
- Thin extremities display red/warm SSS transmission under backlighting.
- Surface retains matte roughness (0.65+) with broad specular highlights.
- Textures track perfectly with skeletal deformation (Zero swimming).

FAILURE MODES [✗]

- Plastic Appearance: IOR/Coat too high, or SSS disabled.
- Scale Breakage: Large procedural fingerprint patterns visible.
- Geometry Melting: Bevel/Subdiv values too high, destroying original KFB character details.
- Noise Flicker: High-frequency fireflies from unfiltered caustics or extreme bump strength.